

Noise Effect Calculator / Noise Profiler

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Noise Effect Calculator for MarMam Detection

Goal: Develop a tool that can be applied to acoustic data and describe the relationship between noise and marine mammal detectability.

Slack: #noise-fx-calc

GitHub: /noise-fx-calc

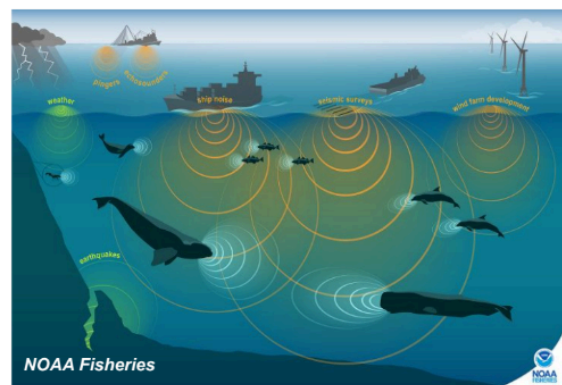
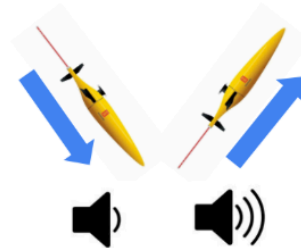
Team members: Will (Aaron) Deans... *All are welcome!*

- *Experienced with:* PAM data, bioacoustics, MATLAB
- **Need experience with:** Glider data/ops, Python, environmental data, statistics?

Potential data sources/tools: Band levels, environmental data, profile data, MarMam detections

Potential challenges: Deciding detection thresholds

First steps: Explore the data!



Sub-proj/second proj?? Noise Profiler

Goal: Develop a tool that can be applied to glider acoustic data and describe noise conditions as a function of glider mode, man-made noise, and environmental variables.

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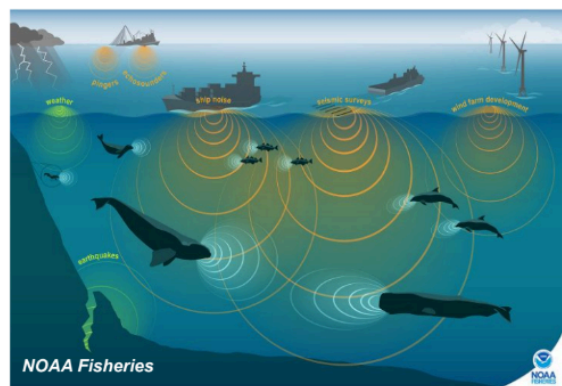
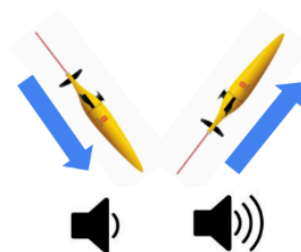
Team members: Will (Aaron) Deans... *All are welcome!*

- *Experienced with:* PAM data, bioacoustics, MATLAB
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Potential data sources/tools: Band levels, environmental data, profile data, *raw acoustic data?*

Potential challenges: Variable instrument sensitivity

First steps: Organize the data!



2 potential parts of this project (can do one/both/split)

- 1) Biological- Idea is for this tool to go through your data & report back to you

Could use someone familiar with gliders, python programmer, environmental data, maybe statistics

- 2) Physical- Noise profiler- understand the glider noise as a function of what the glider is doing

Imagining this as a post-mission application- apply to your data and learn about your data, usability of that data. Are there specific times during your deployment where some acoustic bands might be masked (data not useful).

Consider the idea of machine learning to 'classify' the sounds/noises– but that might be a next step.

The noises the glider makes are pretty stereotyped, so may not require fancy ML. Ambient noise may require fancier processing.

From deep learning perspective, just feeding it more data that contains the type of noise (and it intentionally disregarded) should help with mitigating those stereotyped noises.

Pierre will give talk on Wednesday– he did ask about a glider noise catalog (there is one for seaglider). +1

Maybe the physical might do well before the biological one (to help inform the second).

Starting with getting resources to point out where these tools might exist and where the limitations are (especially as it relates to gliders). +1

Add link to the seaglider noise tool: Grasso et al. 2023 [Quantifying Self-Noise of the Seaglider AUV Using a Passive Acous...: Ingenta Connect](#)

Maybe start with a few examples species on the marine mammal side to start... the impact & method you might use would be different for the two species. Maybe in one case detection threshold/filters might be used, in other cases it might be an effort base (cut out masked sections). Also consider where in the water column you are. +1

-perhaps consider different marine mammal hearing groups as defined by National Marine Fisheries Service

We did a similar experiment few years ago in the Baltic Sea with a G3 Slocum glider. Here the link to the report (not paper) where we describe the noise produced by the glider
<https://julkaisut.valtioneuvosto.fi/server/api/core/bitstreams/0cdecc39-c71e-4e01-b76e-e7a91f596d16/content>

- Thank you Ivia!

Would it be useful to have example noise data

Notes from morning group meeting

Hybrid millidecade data – Kait will include link to github for noise analysis
YAWN

Weds – higher overview

HDF5s should be available

Good to get access to the data first

Julia - familiar with env data and python coding

- Need to scope to what env data we want – What's available?
- Liz +1

Liz - hybrid millidecade will be a straightforward way of getting at some preliminary noise info.
Might miss contribution of marine mammals – confounding var?

Hybrid millidecade does a good job of showing patterns

Would need to factor in how often glider position/orientation data gets updated, and other variables – temporal resolution

Kait: Data are on JupyterHub! (Millidecade data and other resolutions)

Liz – familiar w different platforms; specialize in large data processing; biologist, but learning Python

Julia – Probably our programming person?

Kevin – Not a glider person, active acoustics person (fish).

Active acoustics something to consider for the future... Whether we can put active/passive acoustics devices on same mooring? (If that is even possible)

Kasey (afternoon pitch)- are you hoping this is a predictive or general? General (for now), which will help down the line (eventually help w/ predictive). From selene: It could eventually be used to identify times where the noise is to be expected to be poor for that mission (xx time sucks for fin whales), you might be able to pick & choose your analysis time or surveying time. IT can also play a role in population/detection inferences, and be used for QAQC.

Kait's system can accommodate lat/long. DON't have good ways of visualizing – YAWN

- Required input: Sound file path (Google cloud or local dir)
 - Deployment name
 - Project name (built around fixed sensors)
 - Output location
 - **Optional (below)**
 - Calibration, float or string pointing to CSV with frequency and sensitivity
 - Which channel- default 0
 - Output to day files or full deployments
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